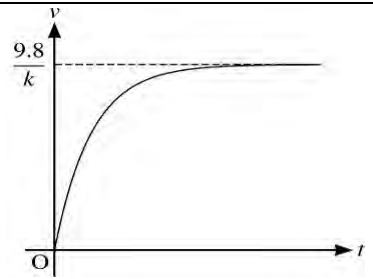
15	**(a)**		$\int \frac{1}{9.8-kv}\,\mathrm{d}v = \int 1\,\mathrm{d}t$ $-\frac{1}{k}\ln(9.8-kv) = t+c$ $9.8-kv = A\mathrm{e}^{-kt}$ When $t=0,\ v=0$ giving $0=\frac{1}{k}(9.8-A\mathrm{e}^{-0})$ $A=9.8$ Hence $v=\frac{9.8}{k}(1-\mathrm{e}^{-kt})$	**M1** **A1** **A1** **M1** **M1** **A1** **A1** **[7]**	**3.1a** **1.1b** **1.1b** **1.1a** **3.1a** **1.1b** **1.1b**	Attempt to separate variables and integrate Correct rearrangement Correctly integrated – must include $+c$ somewhere Attempt to write in the form $v=$ (Clearing natural logs; may be done after evaluation of the constant) Using initial conditions leading to a value for the constant; may be done before clearing logs Correct value for A (or $c=-\frac{1}{k}\ln 9.8$) Must be in the form $v=...$	Allow up to 3M marks for an attempt with an incorrect rearrangement.
			Alternative using a Further Maths method $\frac{\mathrm{d}v}{\mathrm{d}t}+kv=9.8$ Integrating factor $\mathrm{e}^{\int P(t)\,\mathrm{d}t}=\mathrm{e}^{kt}$ $v\mathrm{e}^{kt}=\int\left(9.8\mathrm{e}^{kt}\right)\mathrm{d}t=\frac{9.8}{k}\mathrm{e}^{kt}+c$ $v=\frac{9.8}{k}+c\mathrm{e}^{-kt}$ When $t=0,\ v=0$ giving $c=-\frac{9.8}{k}$ Hence $v=\frac{9.8}{k}(1-\mathrm{e}^{-kt})$	**M1** **A1** **A1** **M1** **M1** **A1** **A1**		Writing in the correct form and using integrating factor Correct integrating factor Correct equation Attempt to write in the form $v=$ Using initial conditions leading to a value for the constant Correct value for c Must be in the form $v=...$	

15	**(b)**			**B1** **B1** **[2]**	**1.1b** **1.1b**	Graph through (0, 0) with the correct general shape or correct for their *v*. Horizontal asymptote labelled $\frac{9.8}{k}$	Ignore any graph for $t < 0$ Any graph involving $\sqrt{t}$ must be concave
15	**(c)**		As $t \to \infty,\ v \to \frac{9.8}{k} = 7$ $k = 1.4$	**M1** **A1** **[2]**	**3.4** **1.1b**	Recognises the limiting value Any form from correct *v*	Allow M1 for their *v* used, provided it has a non-zero limiting value
			Alternative solution Limiting value when $\frac{dv}{dt} = 0$ $k = \frac{9.8}{7} = 1.4$	**M1** **A1**		Could be implied	
15	**(d)**		$v = 3.5 \Rightarrow 1 - e^{-1.4t} = \frac{3.5}{7}$ Hence $-1.4t = \ln 0.5 \Rightarrow t = 0.495$	**B1** **[1]**	**3.4**	May substitute in other versions of the equation relating *v* and *t* Allow for 0.50 or awrt 0.495	No FT from incorrect expression for *v*